In [1]:
```
import pandas as pd

df = pd.read_csv("titanic_train.csv")
df.head()
```

Out[1]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style>

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [2]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [4]:
```
data = df[["Pclass", "Sex", "Age", "Fare", "Survived"]].copy()

data["Sex"] = data["Sex"].map({"male": 0, "female": 1})

data.head()
```

Out[4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Pclass</th> <th>Sex</th> <th>Age</th> <th>Fare</th> <th>Survived</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

3	0	22.0	7.2500	0
1	1	38.0	71.2833	1
3	1	26.0	7.9250	1
1	1	35.0	53.1000	1
3	0	35.0	8.0500	0

In [5]:
```
data["Age"].median()
```

Out[5]: np.float64(28.0)

In [6]:
```
from sklearn.model_selection import train_test_split

X = data[["Pclass", "Sex", "Age", "Fare"]]
y = data["Survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

print("Training:", len(X_train))
print("Testing:", len(X_test))
```

```
Training: 623
Testing: 268
```

In [7]:
```
X_train = X_train.copy()
X_test = X_test.copy()

median_age = X_train["Age"].median()

X_train["Age"] = X_train["Age"].fillna(median_age)
X_test["Age"] = X_test["Age"].fillna(median_age)

print("Missing training ages:", X_train["Age"].isnull().sum())
print("Missing testing ages:", X_test["Age"].isnull().sum())
```

```
Missing training ages: 0
Missing testing ages: 0
```

In [8]:
```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=3, random_state=42)
```

```
model.fit(X_train, y_train)

print("Model trained successfully!")
```

```
Model trained successfully!
```

In [9]:
```
from sklearn.metrics import accuracy_score

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Test accuracy:", round(accuracy, 3))
print("Test accuracy (%):", round(accuracy * 100, 2))
```

```
Test accuracy: 0.806
Test accuracy (%): 80.6
```

In [10]:
```
results = X_test.copy()
results["Actual"] = y_test
results["Predicted"] = y_pred
results["Error"] = results["Actual"] != results["Predicted"]

print("Total mistakes:", results["Error"].sum())

print("\nErrors by class:")
print(results.groupby("Pclass")["Error"].mean())

print("\nErrors by sex:")
print(results.groupby("Sex")["Error"].mean())
```

```
Total mistakes: 52

Errors by class:
Pclass
1    0.259740
2    0.134615
3    0.179856
Name: Error, dtype: float64

Errors by sex:
Sex
0    0.203593
1    0.178218
Name: Error, dtype: float64
```

My model was correct 80.6% of the time. It made 52 mistakes. I noticed that it had the most trouble with first-class passengers.

In [ ]:
```

```